Radiography Registry Mastery

Volume 1 – Physics & Image Production

Chapter 1 – Introduction to Diagnostic Imaging

Draft Notice: This is original educational content intended as a study aid. It is not affiliated with or endorsed by the ARRT.

Learning Objectives

• Describe the purpose of diagnostic imaging.

• Explain the role of the radiographer.

• Identify major imaging modalities.

• Understand why patient care and image quality are equally important.

What Is Diagnostic Imaging?

Diagnostic imaging creates pictures of structures inside the body to assist healthcare providers in diagnosing disease and injury. Radiography uses ionizing radiation to create diagnostic images while balancing image quality with radiation safety.

The Role of the Radiographer

Radiographers verify patient identity, review exam requests, communicate with patients, position anatomy accurately, select exposure factors, produce diagnostic-quality images, evaluate those images, and help maintain patient safety.

Major Imaging Modalities

Radiography: General x-ray imaging.
CT: Cross-sectional x-ray imaging.
MRI: Magnetic fields and radio waves.
Ultrasound: High-frequency sound waves.
Nuclear Medicine: Radioactive tracers to evaluate organ function.

Patient Safety

Always verify the correct patient, explain the examination, protect privacy, screen for pregnancy when appropriate, and follow ALARA principles to minimize radiation exposure.

Clinical Example

A patient presents after a fall with wrist pain. The radiographer confirms the patient's identity, explains the exam, positions the wrist carefully, selects appropriate exposure factors, evaluates the images, and repeats only when necessary.

Memory Aid

RAD SAFE
Review the order
Assess the patient
Determine correct positioning
Shield when appropriate

Apply ALARA
Follow identifiers
Evaluate image quality

Chapter Review Questions

1. What is the primary purpose of diagnostic radiography?

A. Diagnose with images using ionizing radiation (Correct)
B. Perform surgery
C. Prescribe medication
D. Provide physical therapy

2. Which imaging modality does not use ionizing radiation?

A. CT
B. Radiography
C. MRI (Correct)
D. Fluoroscopy

3. Which responsibility belongs to a radiographer?

A. Prescribing medications
B. Producing diagnostic images and positioning patients (Correct)
C. Performing surgery
D. Interpreting all medical images

Chapter Summary

This chapter introduced diagnostic imaging, the responsibilities of the radiographer, common imaging modalities, the importance of patient safety, and foundational concepts that will support the physics and image production chapters that follow.